

Amy Wei

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Education

PhD	University of Massachusetts Amherst , Computer Science • MS/PhD student, advised by Dr. Yuriy Brun	Amherst, MA Sept 2026 – present
BSE	University of Michigan , Computer Science, Minor in Mathematics • GPA: 3.8/4.0, <i>summa cum laude</i> with Engineering Honors • Selected Awards: Barry Goldwater Scholarship	Ann Arbor, MI Aug 2022 – May 2026

Publications

Sweat the Details: Exercise Breaks and AI Assistance in Code Summarization Amy Wei, Priscila Santiesteban, Westley Weimer 10.1109/TSE.2026.3716079 (IEEE Transactions on Software Engineering)	Sept 2026
TestLoop: A Process Model Describing Human-in-the-Loop Software Test Suite Generation Matthew C. Davis, Sangheon Choi, Amy Wei, Sam Estep, Brad A. Myers, Joshua Sunshine 10.1145/3765754 (ACM Transactions on Software Engineering and Methodology)	Aug 2025
TerzoN: Human-in-the-Loop Software Testing with a Composite Oracle Matthew C. Davis, Amy Wei, Brad A. Myers, Joshua Sunshine 10.1145/3729359 (ACM Foundations of Software Engineering)	Apr 2025

Research Experience

University of Michigan , Undergraduate Researcher, Software and Cognition Lab • Advised by Dr. Westley Weimer and PhD student Priscila Santiesteban • Developed paper ideas through literature review of 18 papers • Designed source code stimuli and programming tasks for human study • Conducted study of 47 participants investigating effect of exercise breaks and AI assistance on code summarization performance • Used qualitative analysis and mixed effects linear regression to analyze results • Communicated findings in manuscript	Ann Arbor, MI Nov 2024 – May 2026
Carnegie Mellon University , Research Assistant, REU in Software Engineering '24 • Advised by Dr. Joshua Sunshine, Dr. Brad Myers, and PhD student Matthew Davis • Built TerzoN, a usable testing tool for TypeScript, during 10-week program sponsored by NSF, Amazon, and Meta • Implemented support for 3 types of testing (fuzzing, example-based testing, property-based testing) • Designed user study with 14 participants to evaluate the tool NaNoFuzz against property-based testing tool fast-check for bug-finding performance	Pittsburgh, PA May 2024 – Aug 2024
Carnegie Mellon University , Research Assistant, REU in Software Engineering '23 • Advised by Dr. Joshua Sunshine, Dr. Brad Myers, and PhD student Matthew Davis • Performed heuristic evaluation of 5 testing tools as benchmark (Hypothesis, Jest, Randoop, JQF, Evosuite) • Implemented sorting, human annotation, and validator function features to NaNoFuzz to support multiple modes of testing • Presented at weekly meetings and shared findings in department-wide poster session with 100+ attendees	Pittsburgh, PA May 2023 – Aug 2023

Teaching

University of Michigan , Instructional Aide, Discrete Mathematics (EECS 203)	Ann Arbor, MI
- Served as a TA for 5 semesters, led homework management team as czar for 2 semesters - Taught discussion section of 40+ students to reinforce course concepts in a collaborative environment - Held weekly office hours to support students in proof-based homeworks - Hired and managed team of 16 graders to ensure students received accurate and timely feedback - Led end-to-end homework duties, published homework grades for 600+ students - Developed supplemental module of 33 problems to help students build mathematical proof-writing skills	Jan 2024 – May 2026

Honors and Awards

Barry Goldwater Scholarship – \$7,500 , Barry Goldwater Foundation	Mar 2025
The preeminent undergraduate award in STEM. Awarded for excellence in research, academics, and contributions to the community. 50 scholars in mathematics and computer science were selected nationally in 2025.	
College of Engineering Scholarship Award – \$3,000 , U-M College of Engineering	Aug 2025
Arlen R. Hellwarth Award – \$500 , U-M College of Engineering	Feb 2025
College of Engineering Scholarship Award – \$2,000 , U-M College of Engineering	Aug 2024
Distinguished Leadership Award , U-M College of Engineering	Feb 2024
George Landes Prize, Technical Communication First Year – \$150 , University of Michigan	Mar 2024
William J. Branstrom Freshman Prize , University of Michigan (top 5% of class)	Feb 2023

Activities

Goldwater Scholar Council – Mentorship Committee, Alumni Taskforce Committee	Aug 2026 – present
- Interviewed Goldwater Scholar alumni to improve future engagement opportunities for the community - Organized 3 month mentorship program to support researchers' academic and personal development	
IEEE-Eta Kappa Nu (EECS Honor Society)	Sept 2023 – May 2026
Earned highest number of service hours in University of Michigan chapter	
Society of Women Engineers	Sept 2022 – May 2026
- Volunteered at Elementary Engineering Exploration Day to introduce students to STEM - Held a charity Fun Run in support of U-M Health C.S. Mott Children's Hospital	
Campus Symphony Orchestra, Violin	Sept 2022 – Apr 2026
Rehearsed weekly in auditioned ensemble, performed in university-wide concerts each semester as Violin I and Principal Violin II	
Applied Linear Algebra (MATH 214), Grader	Sept 2023 – Jan 2024
- Graded weekly homework for 300+ students, drafted comprehensive grading rubrics - Wrote comments to provide students with individualized feedback	
Women's Glee Club Choir, Alto I	Sept 2022 – Apr 2023
Rehearsed weekly in auditioned ensemble, performed in university-wide concerts	

Skills

Technical: C/C++, Python, JavaScript/TypeScript, Java, Scheme, R, HTML/CSS, Git, LaTeX